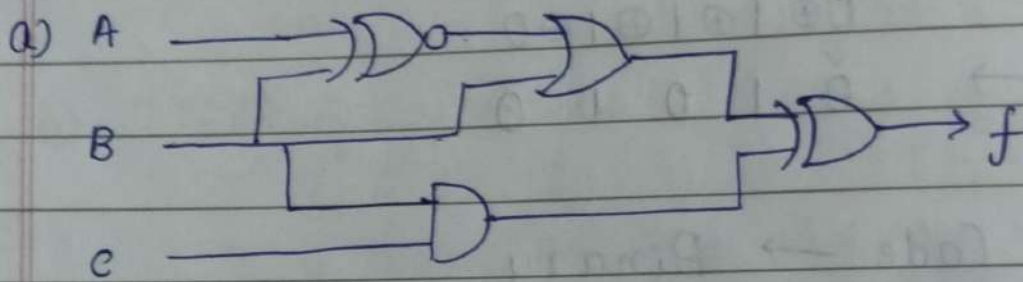


Assignment - 2 Boolean Algebra

1) Derive the output expression 'f' for the below circuits and draw the circuits



$$Z \Rightarrow (A+B)' + B$$

$$Y = BC$$

$$= AB + \bar{A}\bar{B} + B$$

$$= B(A+1) + \bar{A}\bar{B}$$

$$= B + \bar{A}\bar{B}$$

$$Z = \underline{\underline{\bar{A} + B}}$$

$$\therefore f = Z \oplus Y$$

$$= (\bar{A} + B) \oplus BC$$

$$= (\bar{A} + B)(\overline{BC}) + (\bar{A} + B)BC$$

$$= (\bar{A} + B)(\bar{B}\bar{C}) + (\bar{A} + B)BC \quad (BB=B)$$

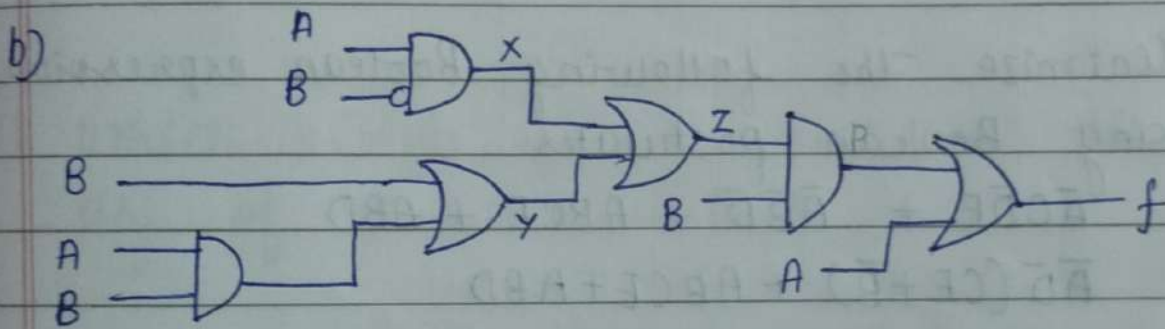
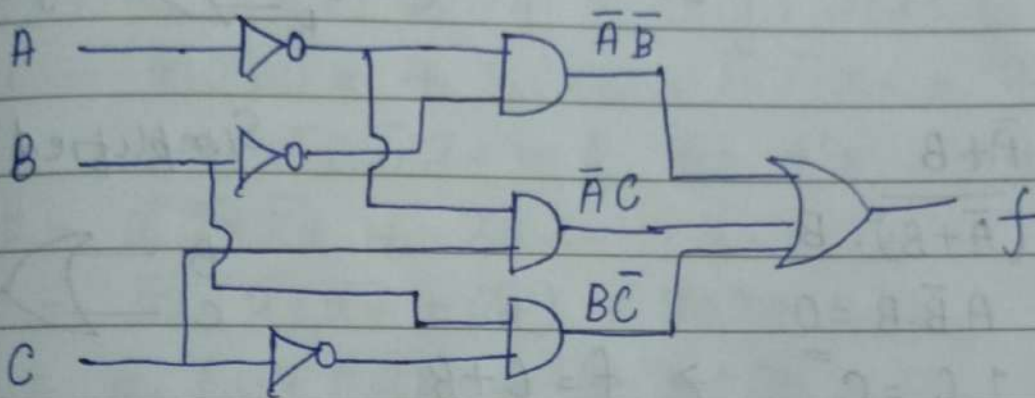
$$= (\bar{A} + B)(\bar{B}\bar{C})$$

$$= (\bar{A} + B)(\bar{B} + \bar{C})$$

$$= \bar{A}\bar{B} + \bar{A}\bar{C} + B\bar{B} + B\bar{C}$$

$$f = \underline{\underline{\bar{A}\bar{B} + \bar{A}\bar{C} + B\bar{C}}}$$

Simplified Circuit :



$$X = A\bar{B}$$

$$Y = B + AB = B(1 + A) = B$$

$$Z = X + Y$$

$$= A\bar{B} + B$$

$$= A + B$$

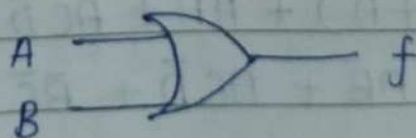
$$P = ZB$$

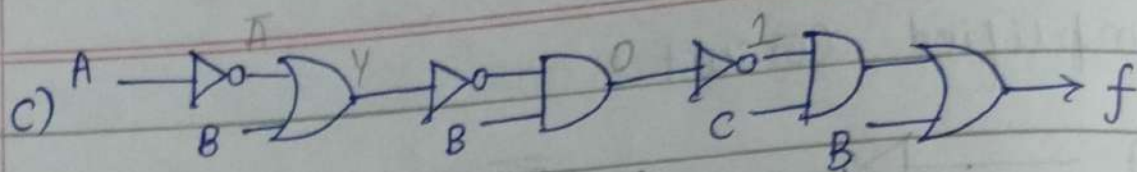
$$= (A + B)B = AB + BB = AB + B = B$$

$$f = A + P$$

$$= A + B$$

∴ Simplified
Circuit :





Simplified circuit

$$Y = \bar{A} + B$$

$$Z = (\bar{A} + B) \cdot B$$

$$= A\bar{B} \cdot B = 0$$

$$1 = 1 \cdot C = C \Rightarrow f = \underline{\underline{C + B}}$$



27 Minimize the following Boolean expressions using Boolean postulates.

a) $\bar{A}\bar{C}\bar{D}E + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

$$\bar{A}\bar{D}(CE + \bar{B}) + ABCE + ABD$$

$$F = \bar{A}\bar{D}(CE + \bar{B}) + AB(CE + D)$$

=

b) $\bar{A}\bar{B}\bar{C} + ABD + \bar{A}\bar{C} + \bar{A}\bar{C}\bar{D} + A\bar{C}\bar{D} + A\bar{B}\bar{C}$

$$\bar{A}\bar{B}\bar{C} + ABD + \bar{A}\bar{C}(1 + \bar{D}) + A\bar{C}\bar{D} + A\bar{B}\bar{C}$$

$$\bar{A}\bar{B}\bar{C} + ABD + \bar{A}\bar{C} + A\bar{C}\bar{D} + A\bar{B}\bar{C}$$

$$\bar{B}\bar{C}(\bar{A} + A) + ABD + \bar{A}\bar{C} + A\bar{C}\bar{D}$$

$$F = \bar{B}\bar{C} + \bar{A}\bar{C} + ABD + A\bar{C}\bar{D}$$

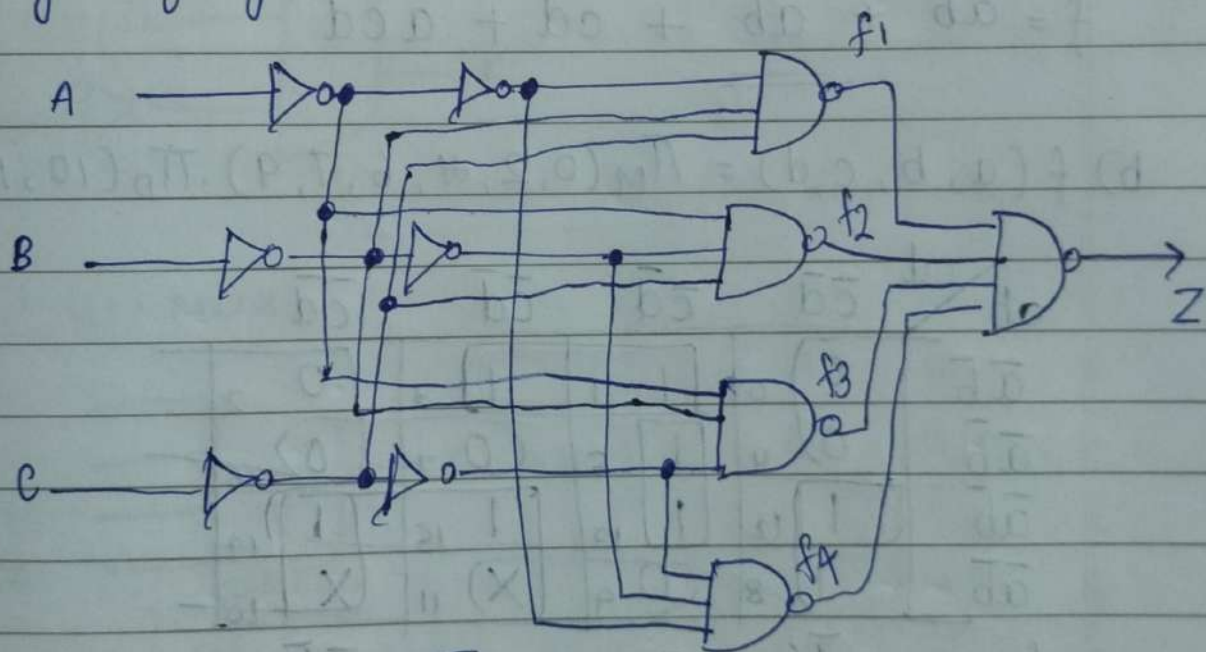
c) $\bar{A}\bar{C} + BC + A\bar{B} + \bar{A}BC + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$

$$\bar{A}\bar{C} + BC(1 + \bar{A}) + A\bar{B} + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$F = \bar{A}\bar{C} + BC + A\bar{B} + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$$

d) $\bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_1 x_2 \bar{x}_3 x_4 + x_2$
 $= x_2 + \bar{x}_1 \bar{x}_2 \bar{x}_3 x_4 = x_2 (1 + \bar{x}_1 \bar{x}_2 \bar{x}_3 x_4) = x_2$
 $F = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_2$
 $x_2 + \bar{x}_2 \bar{x}_1 x_4 \quad (A + A'B = A + B) = x_2 + \bar{x}_1 x_4$
 $F = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + x_2 + \bar{x}_1 x_4$
 $= \bar{x}_1 (\bar{x}_3 \bar{x}_4 + x_4) + x_3 x_4 + x_2$
 $= \bar{x}_1 (x_4 + \bar{x}_3) + x_3 x_4 + x_2$
 $F = \bar{x}_1 x_4 + \bar{x}_1 \bar{x}_3 + x_3 x_4 + x_2$

3) Which single gate can represent the functionality of given circuit?



$$f_1 = \overline{ABC} \quad f_2 = \overline{AB\bar{C}} \quad f_3 = \overline{\bar{A}BC} \quad f_4 = \overline{\bar{A}\bar{B}\bar{C}}$$

$$= \bar{A} + \bar{B} + C \quad = A + \bar{B} + C \quad = A + B + \bar{C} \quad = \bar{A} + \bar{B} + \bar{C}$$

$$Y = [\bar{A} + \bar{B} + C \cdot A + \bar{B} + C \cdot A + B + \bar{C} \cdot \bar{A} + \bar{B} + \bar{C}]$$

$$= \bar{A}\bar{B}\bar{C} + \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC = A \oplus B \oplus C \quad \underline{\underline{\text{XOR Gate}}}$$

4. Minimize the following Boolean Expressions using K-map

a) $f(a, b, c, d) = \sum m(4, 6, 8, 10, 11, 12, 15) + \sum d(3, 5, 7, 9)$

$ab \backslash cd$	$\bar{c}\bar{d}$	$\bar{c}d$	$c\bar{d}$	cd
$\bar{a}\bar{b}$	0	1	X	3
$\bar{a}b$	1	X	X	7
ab	1	1	1	15
$a\bar{b}$	1	X	1	11

$$f = \bar{a}b + a\bar{b} + cd + a\bar{c}\bar{d}$$

b) $f(a, b, c, d) = \prod M(0, 2, 4, 6, 7, 9) \cdot \prod D(10, 11)$

$ab \backslash cd$	$\bar{c}\bar{d}$	$\bar{c}d$	$c\bar{d}$	cd
$\bar{a}\bar{b}$	0	1	1	3
$\bar{a}b$	0	1	0	7
$a\bar{b}$	1	1	1	15
ab	1	0	X	X

$$f = (a+d) \cdot (a\bar{b} + \bar{c}) \cdot (\bar{a} + b + \bar{d})$$